

World Conference on Transport Research - WCTR 2026 Toulouse 6-10 July 2026

PathFlowQuery-GNN: A Spatio-Temporal Learning Approach for Urban Freight Path Flow Estimation

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Abstract

Rapid growth in urban freight demand has amplified the impact of trucks on urban traffic systems, yet most management strategies remain passenger-oriented. Progress has been limited by sparse AVI deployment and biased, low-sampling GPS data, which hinder reliable estimation of freight movements. To address this challenge, we propose a calibrated multi-source data fusion pipeline and a novel Path Flow Query Graph Neural Network (PFQ-GNN) that directly maps AVI detector streams to path-level freight flows. The model integrates edge-wise temporal encoding, efficient edge-to-edge message passing, and regularized attention-based path readout, enabling scalable and interpretable flow estimation without explicit path enumeration. On a large-scale urban freight network, PFQ-GNN consistently outperforms a strong 3D-CNN baseline in both absolute and percentage errors, while producing corridor-level attention patterns that reveal dominant OD pairs and route choices. The results provide valuable information and a critical reference for the development of effective urban freight vehicle traffic management policies.

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Peer-review under responsibility of the scientific committee of the World Conference on Transport Research - WCTR 2026.

Keywords: Urban Freight; Path Flow Estimation; Spatio-temporal Graph Neural Network; Data Fusion.

1. Introduction

In recent years, driven by the rapid expansion of e-commerce and intra-city delivery services, the reconfiguration of industrial spatial patterns, and the compression of nighttime logistics windows, urban freight demand has exhibited sustained and substantial growth. Freight vehicles differ markedly from passenger cars and public transit vehicles in axle load, vehicle dimensions, and operating time windows; consequently, they exert disproportionately large marginal effects on roadway capacity, traffic variability, and environmental externalities such as congestion, pollutant emissions, and noise. Moreover, the spatial spillover effects of warehousing and distribution facilities have amplified the localized traffic and environmental burdens associated with heavy-duty vehicle activity. Empirical evidence indicates that warehousing expansion is associated with notable increases in near-surface NO₂ concentrations in adjacent neighbor-

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hoods, underscoring the coupled traffic–environment implications of door-to-door consumption models at the urban scale [Stokenberga et al. \(2023\)](#); [Kerr et al. \(2024\)](#).

Despite these trends, extant traffic management and demand-side policies remain predominantly passenger-centric, and there is a persistent lack of systematic understanding of freight vehicle travel characteristics and microscopic behaviors—particularly with respect to route choice, temporal distribution, and vehicle-type heterogeneity. Reliance on link-level traffic counts and aggregate indicators is increasingly insufficient for fine-grained governance. Effective urban freight management therefore requires estimation frameworks that are explicitly oriented to time-varying origin–destination (OD) demand and path-level flows, enabling the characterization of demand generation and transfer in space and time and elucidating how multi-path allocation shapes the spatial dispersion and temporal concentration of network load [Federal Highway Administration \(2012\)](#); [Transportation Research Board \(2010\)](#).

Methodologically, AVI’s vehicle-level timestamps and identity chains can be used to register and filter GPS trajectories, thereby reducing matching errors and enabling calibration of time-varying travel times, congestion thresholds, and compliance/avoidance behavioral parameters. At the same time, a large body of GPS-based freight behavior research—spanning trip-end/stop identification and route-choice modeling—demonstrates the value of GPS for revealing delivery stops and corridor preferences; yet, when used as a **single source**, GPS suffers from penetration and sampling bias (e.g., vendor/firm participation, vehicle class mix), threshold sensitivity and drift in stop detection, and limited representativeness across time and space, which can propagate bias into activity and route elasticity estimates and hinder network-level, path-flow inference required for policy evaluation [Mufti et al. \(2021\)](#); [Yang et al. \(2022\)](#). At the inferential level, GPS-derived candidate path sets and prior route-share distributions for OD pairs, when coupled with AVI’s cross-point temporal consistency constraints in a dynamic estimation framework, mitigate the ill-posedness of path-level flow inference and enhance extrapolation to weakly observed links and sparsely sampled periods. Prior studies confirm that integrating AVI with probe-/connected-vehicle trajectories materially improves the accuracy and robustness of dynamic OD and path-flow estimation at city scale, underscoring the necessity of multi-source fusion beyond single-source GPS.

1.1. Literature Review

In order to support fine-grained governance of urban freight systems, the research focus has progressively shifted from link-volume–based indirect inference toward integrated estimation of dynamic Origin–Destination (OD) demand and path-level flows using multi-source observations. The literature in OD/path-flow estimation can be broadly categorized into two methodological families: non-learning approaches and learning-based approaches.

Non-learning approaches to OD and path flow estimation mainly rely on optimization or statistical inference frameworks that infer flows indirectly from traffic counts, AVI/ANPR over-point identifications, or other aggregate observations. Early static or semi-dynamic models employed generalized least squares, entropy maximization, or Bayesian inference under equilibrium or prior constraints, producing interpretable parameters but typically assuming fixed route assignments and overlooking time-varying route choice behavior [Yang and Huang \(1992\)](#); [Cascetta and Posterino \(1993\)](#). To capture temporal evolution, state-space and filtering methods such as Kalman filters, extended Kalman filters, and particle filters were developed, providing improved handling of daily or intra-day variation, although they remain sensitive to noise, sparse sensor coverage, and identifiability issues [Ashok and Ben-Akiva \(2002\)](#). Another important stream of work integrates AVI/ANPR over-point identity information with link counts to enhance identifiability; representative studies include van der Zijpp’s joint estimation of OD demand using AVI and counts [van der Zijpp \(1997\)](#), Dixon and Rilett’s fusion of generalized least squares with filtering [Dixon and Rilett \(2002\)](#), and Zhou and Mahmassani’s AVI-based dynamic OD estimation framework [Zhou and Mahmassani \(2006\)](#). More recently, assignment-free or approximated dual-level formulations have been proposed to reduce the computational burden of dynamic traffic assignment, where the upper level estimates OD demand while the lower level applies simplified or relaxed assignment models, albeit at the cost of behavioral realism and path-level granularity [Ros Roca et al. \(2022\)](#).

Learning-based approaches have increasingly been applied to OD and path flow estimation, leveraging machine and deep learning models to capture complex, often non-linear, spatio-temporal dependencies. A representative line of work employs tensorized AVI observations (organized by space, time, and detector) as inputs to three-dimensional convolutional neural networks (3D CNNs, e.g., Res3D) to infer dynamic OD demand, often supplementing limited real data with simulation-generated samples to alleviate label scarcity [Tang et al. \(2021\)](#). Other studies focus on integrating

GPS or connected-vehicle trajectory data with AVI/ANPR observations: these approaches construct candidate path sets, prior route shares, or virtual AVI points and then apply hybrid deep learning models—such as 3D CNNs with feature fusion and noise calibration—for robust path flow estimation [Chen et al. \(2021\)](#); [Cao et al. \(2021\)](#); [Ma et al. \(2022\)](#).

More recently, semi-supervised learning frameworks based on graph neural networks (GNNs) have been proposed, often combined with generative adversarial networks (GANs) or consistency regularization strategies, to enforce distributional consistency across both observed and unobserved OD pairs under conditions of low AVI or probe-vehicle penetration. For instance, [Zhang et al. \(2024\)](#) develop a semi-supervised arterial path flow estimation model that fuses sparse AVI data and low penetration connected-vehicle (CV) trajectories using multi-knowledge graphs, and deploy a GAN discriminator to ensure that estimated path-flow distributions align with observed proportions even when only a small fraction of OD pairs are observed. Another line of work investigates dynamic OD estimation using probe-vehicle or GPS-only learning methods: [Cao et al. \(2024\)](#) propose OD flow estimation models that depend exclusively on GPS trajectories, achieving fine temporal resolution; however, these methods remain constrained by limited spatial coverage, sampling bias among OD pairs, and reduced generalization to under-observed links or times. Collectively, these advances underscore the potential of semi-supervised and probe-only approaches—but also highlight recurring limitations in path enumeration, label noise, and weak-coverage external validity.

Recent advances in deep learning for traffic forecasting have evolved from diffusion or graph convolution with recurrent sequence modeling [Li et al. \(2018\)](#), to fully convolutional spatiotemporal architectures [Yu et al. \(2018\)](#), and more recently to adaptive and attention-driven mechanisms [Wu et al. \(2019\)](#); [Guo et al. \(2019\)](#); [Zheng et al. \(2020\)](#). These developments have progressively enhanced the ability to capture long-range dependencies, multi-scale temporal dynamics, and adaptive spatial correlations in transportation networks. However, most existing models remain node- or link-centric, focusing on traffic states at fixed locations rather than path-level dynamics, and often require explicit path enumeration, which limits scalability and interpretability in large urban freight networks. Building on this foundation, the present study introduces an edge-centric spatiotemporal architecture with a learnable query-based readout, enabling direct path-flow estimation without explicit enumeration and thus better supporting policy-oriented analysis under realistic data constraints.

1.2. Contribution of This Paper

To fill the aforementioned research gaps, this study proposes a city-scale, dynamically adaptive framework for path-level freight flow estimation that integrates data fusion, simulation-based augmentation, and spatio-temporal graph learning. The key contributions are:

- **Multi-source data fusion with simulation-driven augmentation.** We fuse *real* truck GPS trajectories with AVI detector records to construct freight OD–path priors and to *calibrate* SUMO inputs. By incorporating domain-randomized simulation as an augmentation mechanism, the framework alleviates label scarcity and data bias while remaining consistent with detector coverage.
- **PFQ-GNN for detector→path mapping.** We introduce a compact ST-GNN that *directly* estimates path flows from spatio-temporal detector data, fully leveraging AVI observations rather than aggregating to link-level flows. The architecture combines edge-wise dilated temporal convolutions with on-graph edge-to-edge propagation, and a query-based soft-path readout, enabling efficient and scalable path-level flow estimation in large urban networks.
- **Attention-based interpretability.** We incorporate entropy and Laplacian based regularization that improves the interpretability of the soft-path attention mechanism, producing corridor-level flow patterns that align with observed freight movements. This provides transparent, explainable insights that support freight corridor management and policy analysis.

2. Preliminaries

This section introduces the fundamental definitions, notations, and problem statement that underpin the proposed framework. We describe the representation of the road network, the characteristics of AVI and GPS data sources, and formally define the path flow estimation problem.

2.1. Network Representation

Let the urban road network be modeled as a directed graph $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of intersections or AVI detector locations ($|\mathcal{V}| = N$), and $\mathcal{E}$ is the set of directed road segments ($|\mathcal{E}| = E$). Each edge $e = (i \rightarrow j) \in \mathcal{E}$ has an associated free-flow travel time τ_e and length l_e .

The study horizon is discretized into uniform time steps $\mathcal{K} = \{1, \dots, K\}$ with resolution Δt . We assume $\Delta t \gg \tau_e$ so that vehicle transitions within a time step can be treated as synchronous.

A path p is defined as an ordered sequence of connected edges $p = (e_1, e_2, \dots, e_m)$ such that $e_k = (v_{k-1} \rightarrow v_k) \in \mathcal{E}$. Let $\mathcal{P}$ denote the set of candidate freight paths, with $|\mathcal{P}| = P$. In the experimental setting, $\mathcal{P}$ is partitioned into heavy-duty ($\mathcal{P}^H$) and light-duty ($\mathcal{P}^L$) subsets, reflecting heterogeneous vehicle behaviors.

2.2. AVI and GPS Observations

AVI detectors. Automatic Vehicle Identification (AVI, including ETC/ANPR systems) provides high-frequency point observations with vehicle identifiers. For detector node $i \in \mathcal{V}$ at time step k , the node feature $x_i(k)$ represents the number of identified vehicles. For a directed detector link $(i \rightarrow j) \in \mathcal{E}$, the edge feature $z_{ij}(k) = [M_{ij}(k), \tau_{ij}(k)]^\top$ contains the matched count $M_{ij}(k)$ and observed travel time $\tau_{ij}(k)$. If no match occurs, $z_{ij}(k) = \mathbf{0}$.

GPS trajectories. GPS data from heavy-duty trucks provide continuous spatiotemporal traces. After map-matching using a Hidden Markov Model (HMM), each trajectory is represented as a sequence of traversed edges. The first and last nodes define the corresponding OD pair. Compared to AVI, GPS offers route visibility but suffers from low penetration and sampling bias.

2.3. Problem Statement

The goal of this study is to estimate time-varying *path-level flows* for freight vehicles, leveraging AVI and GPS observations.

For a given window $\mathcal{W}_t = \{t - T + 1, \dots, t\}$ of length T , let the sequence of attributed detector graphs be denoted as $\mathcal{G}_t = (G_k)_{k \in \mathcal{W}_t}$. The supervised target is a P -dimensional vector of path flows:

$$\mathbf{y}(t) = [y_1(t), \dots, y_P(t)]^\top, \quad y_p(t) = \sum_{k \in \mathcal{W}_t} c_p(k), \quad (1)$$

where $c_p(k)$ is the number of vehicles traversing path p at time step k .

The estimation problem is thus formulated as learning a mapping

$$f_\theta : \mathcal{G}_t \mapsto \hat{\mathbf{y}}(t), \quad (2)$$

where f_θ is a parametric model with parameters θ , trained to minimize the expected loss $\ell(\hat{\mathbf{y}}(t), \mathbf{y}(t))$ under non-negativity, temporal smoothness, and flow-conservation regularization.

This formulation highlights the inherent challenges: (i) the dimensional mismatch between detector-level observations ($N + E$ signals per step) and path-level outputs (P flows), leading to ill-posedness; (ii) the irregular and directed graph topology; and (iii) the multi-scale temporal dependencies of traffic dynamics. These motivate the adoption of

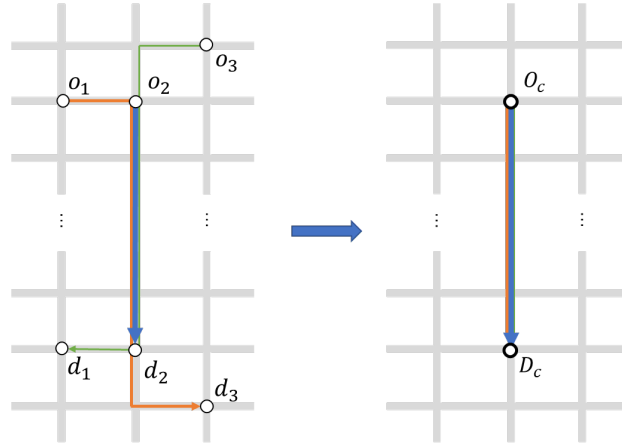


Fig. 1. OD Clustering

spatio-temporal graph neural networks, which naturally capture non-Euclidean spatial structures and temporal correlations.

3. Methodology

3.1. Data Processing Framework

This study develops a data processing pipeline that fuses GPS trajectory data from heavy-duty trucks with AVI detector observations in order to construct robust, simulation-ready inputs for path flow estimation. The framework consists of three stages: (i) OD–path set construction, (ii) prior path flow construction, and (iii) simulation-based data augmentation. Together, these steps transform raw observational data into structured inputs that preserve behavioral realism while remaining computationally tractable.

3.1.1. OD–Path Set Construction

We begin by extracting origin–destination (OD) pairs and detailed travel paths of individual freight vehicles from GPS trajectories. Each trajectory is aligned to the underlying road network using a Hidden Markov Model (HMM)-based map-matching algorithm, producing a complete sequence of traversed road segments. The first and last nodes of each trajectory define its OD pair, capturing both trip demand and route choice behavior.

Large-scale urban freight networks typically yield a prohibitively large number of OD pairs, many of which exhibit strong spatial similarity. To reduce complexity while preserving network topology, OD-pair similarity is defined by the sum of shortest-path distances between corresponding origins and destinations:

$$\text{dist}_{OD}(i, j) = \text{sp}(o_i, o_j) + \text{sp}(d_i, d_j), \quad (3)$$

where $\text{sp}(\cdot, \cdot)$ is the directed shortest-path distance computed on the road network G via Dijkstra's algorithm. The resulting distance matrix is embedded into a low-dimensional Euclidean space (e.g., 4D) using Multidimensional Scaling (MDS), followed by KMeans clustering. Each cluster $\{C_k\}$ is represented by its most frequent and spatially central OD pair, thereby yielding a reduced and tractable OD set while maintaining spatial representativeness.

To account for the greater spatial flexibility of light-duty freight vehicles, the OD set is further enriched by incorporating freight-related Points of Interest (POIs), such as factories, ports, and logistics parks. Road network access points to these POIs are treated as candidate OD terminals and spatially aligned with nearby OD clusters before being added to the final OD set. This ensures comprehensive coverage of heterogeneous freight vehicle types.

For each OD pair, a diverse set of candidate paths is generated using a penalized shortest-path algorithm. Starting from the standard shortest path, multiplicative penalties are applied iteratively to previously used edges:

$$w'_e = \lambda \cdot w_e, \quad \lambda > 1, \quad (4)$$

with the search repeated until an adaptive upper bound K (up to 20 paths) is reached. Candidate paths are filtered to exclude those with excessive overlap or large detour ratios, ensuring diversity and practicality. Metadata such as path length, number of segments, and arterial road proportion are also recorded. The resulting OD–path set $\{P_{od}\}$ balances topological diversity with manageable scale, serving as the basis for both AVI-based inference and simulation inputs.

3.1.2. Prior Path Flow Construction

Given the OD–path set as a “dictionary,” we construct time-varying prior estimates of path-level flows by aggregating GPS trajectories and AVI detector records within sliding time windows. The objective is to match incomplete observational evidence to candidate paths in a probabilistic manner.

For each GPS trajectory γ (treated as ground truth after HMM map-matching), we compute its similarity to a candidate path p via an overlap coefficient:

$$C_{p,\gamma} = \frac{l_{p,\gamma}}{\sqrt{L_p L_\gamma}}, \quad (5)$$

where L_p and L_γ denote the lengths of the candidate and observed paths, and $l_{p,\gamma}$ is the total length of their overlapping segments. Aggregating over all trajectories in a time window yields heavy-duty vehicle prior flows $f_p^{\text{GPS}}(t)$. AVI records, while not full trajectories, are treated as low-frequency “partial traces.” Matching observed detector sequences against candidate paths yields corresponding light-duty priors $f_p^{\text{AVI}}(t)$.

These priors encode realistic spatiotemporal demand patterns while allowing for uncertainty. To increase robustness, we aggregate one week of empirical data to construct daily priors, and then perturb each hourly flow with Gaussian noise proportional to its magnitude:

$$\tilde{f}_p(t) = f_p(t) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 f_p(t)^2), \quad \sigma = 0.1. \quad (6)$$

3.1.3. Simulation-Based Augmentation

The noisy prior path flows $\{\tilde{f}_p(t)\}$ serve as inputs to the SUMO microsimulation platform, configured with the real urban road network and consistent AVI detector placement. During simulation, only detector-level time series (counts, matched volumes, travel times) are retained as inputs to the learning model, while path-level information is withheld to prevent leakage. This design ensures that the model learns to recover path flows solely from detector observations, mirroring real-world deployment conditions.

The resulting synthetic samples inherit the spatial and temporal patterns of observed freight traffic while introducing controlled perturbations that enhance coverage. These augmented datasets provide the foundation for training a spatio-temporal graph neural network to map AVI observations to path-level flows.

3.2. Model Formulation

The AVI observation system is represented as a directed detector graph $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of detector nodes and directed links, with cardinalities $N = |\mathcal{V}|$ and $E = |\mathcal{E}|$, respectively (Fig. 5). The analysis horizon is discretized into uniform intervals $\mathcal{K} = \{1, \dots, K\}$ of length Δt . Let τ_e be the mean travel time on edge $e \in \mathcal{E}$. Assuming $\Delta t \gg \tau_e$, vehicle movements within a step can be regarded as synchronous. At each time step k ,

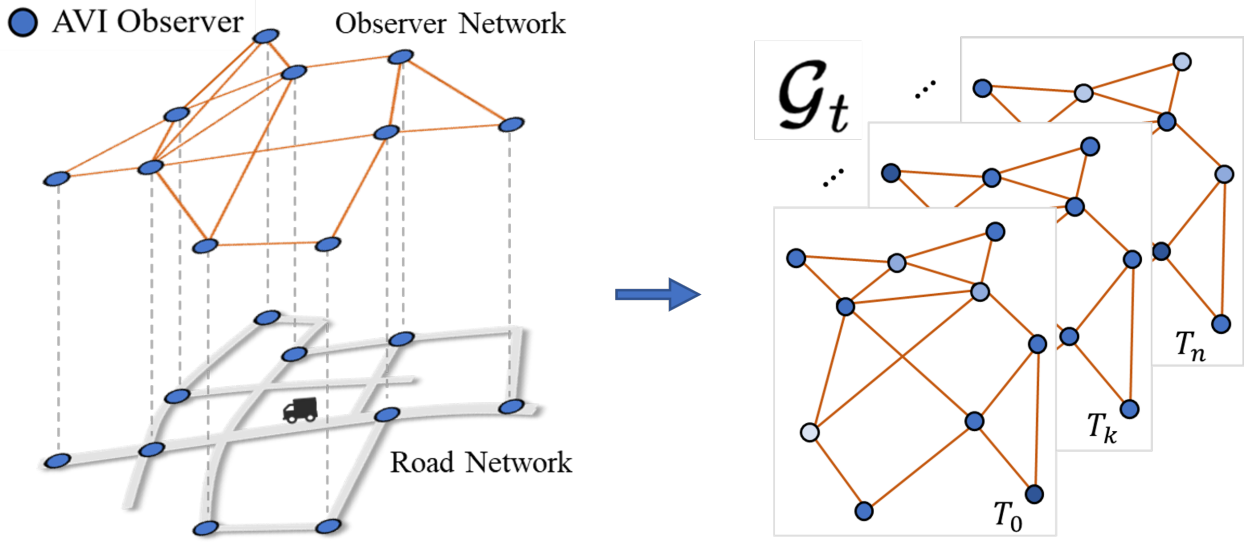


Fig. 2. Detector-based graph representation of AVI observations

detector node $i \in \mathcal{V}$ records a nonnegative vehicle count $x_i(k)$. For each directed link $(i, j) \in \mathcal{E}$, the edge feature is $z_{ij}(k) = [M_{ij}(k), \tau_{ij}(k)]^\top$, where $M_{ij}(k)$ denotes the matched volume and $\tau_{ij}(k)$ the observed travel time. If no match occurs, the edge attribute is set to zero.

For a temporal window of length T , we define the index set $\mathcal{W}_t = \{t - T + 1, \dots, t\}$ and the corresponding attributed graph sequence $\mathcal{G}_t = (G_k)_{k \in \mathcal{W}_t}$. The graph structure remains fixed, while node and edge features evolve dynamically. Importantly, all features are derived solely from AVI observations, without access to ground-truth path flows, thereby preventing information leakage. To address sparsity and scale heterogeneity, only nonzero entries are normalized (e.g., using $\log(1 + x)$ combined with quantile scaling), leaving zeros unchanged.

Graph and time grid: $G = (\mathcal{V}, \mathcal{E})$, $N = |\mathcal{V}|$, $E = |\mathcal{E}|$, $\mathcal{K} = \{1, \dots, K\}$, $\Delta t \gg \tau_e (\forall e)$,

Node features: $X^V(k) = [x_i(k)]_{i \in \mathcal{V}} \in \mathbb{R}^{N \times 1}$, $x_i(k) \geq 0$,

Edge features: $X^E(k) = [z_{ij}(k)]_{(i,j) \in \mathcal{E}} \in \mathbb{R}^{E \times 2}$, $z_{ij}(k) = [M_{ij}(k), \tau_{ij}(k)]^\top$,

Windowed input: $\mathcal{W}_t = \{t - T + 1, \dots, t\}$, $\mathcal{G}_t = (G_k)_{k \in \mathcal{W}_t}$.

(7)

Let $\mathcal{P} = \mathcal{P}^H \cup \mathcal{P}^L$ denote the target path set, partitioned into heavy-duty ($\mathcal{P}^H$) and light-duty ($\mathcal{P}^L$) subsets, with sizes $P_H = |\mathcal{P}^H|$, $P_L = |\mathcal{P}^L|$, and $P = P_H + P_L$. In experiments, we take $P_H = P_L = 100$, hence $P = 200$. For each path $p \in \mathcal{P}$, the aggregated flow within window $\mathcal{W}_t$ is defined as the total number of vehicles traversing p , yielding the path flow vector:

Path sets: $\mathcal{P} = \mathcal{P}^H \cup \mathcal{P}^L$, $P_H = |\mathcal{P}^H|$, $P_L = |\mathcal{P}^L|$, $P = P_H + P_L$,

Windowed path flow: $y_p(t) = \sum_{k \in \mathcal{W}_t} c_p(k) \geq 0$, $p \in \mathcal{P}$,

Target vector: $\mathbf{y}(t) = [y_1(t), \dots, y_P(t)]^\top \in \mathbb{R}^P$.

(8)

The learning problem is thus to infer the mapping from AVI-derived graph sequences to path-level flows. Let f_θ denote the parametric model with parameters θ . The training objective minimizes the expected loss between predicted and true path flows, with optional regularization terms (e.g., non-negativity, temporal smoothness, and conservation constraints):

$$\begin{aligned} \text{Model mapping: } f_\theta : \mathcal{G}_t &\mapsto \hat{\mathbf{y}}(t) \in \mathbb{R}^P, \\ \text{Objective: } \theta^\star &= \arg \min_\theta \mathbb{E}[\ell(\hat{\mathbf{y}}(t), \mathbf{y}(t))]. \end{aligned} \quad (9)$$

This formulation highlights the core challenge of AVI-based path flow estimation: transforming localized, detector-level signals into consistent path-level flows. Detector counts provide node-level observations, while matched volumes and travel times provide directional edge features, together forming a sequence of attributed directed graphs. However, a dimensional gap arises: the number of detector signals ($N + E$) is typically far smaller than the number of path flows (P), making the problem underdetermined. Moreover, the underlying road network is non-Euclidean, directed, and structurally imbalanced, while congestion dynamics evolve across multiple temporal scales.

These characteristics motivate the adoption of spatio-temporal graph learning. Rather than relying on Euclidean assumptions, spatio-temporal graph neural networks (ST-GNNs) can model how flows propagate along directed edges and interact dynamically over time. By embedding AVI observations into this framework, the proposed formulation bridges the gap between sparse detector evidence and high-dimensional path flows, enabling a scalable and interpretable solution to the AVI-based OD estimation problem.

3.3. Model Architecture

Edge–Time Encoding

For each directed link ($i \rightarrow j$) and step $k \in \mathcal{W}_t$, we construct an *edge-centric, temporally contextual* representation that (i) respects directionality and edge attributes at the same step, and (ii) captures multi-scale dependencies across the window. We realize this via a *same-step, edge-conditioned gate* followed by a *dilated 1-D temporal convolution (TCN)*; the former places edge labels directly in the message function (edge-conditioned filters in the MPNN view), while the latter supplies long effective memory with parallel training (Simonovsky and Komodakis, 2017; Gilmer et al., 2017; Bai et al., 2018; van den Oord et al., 2016; Wu et al., 2019).

The source detector count is projected to a hidden vector, and the edge attributes generate a gate that modulates the source signal:

$$\mathbf{h}_i(k) = \sigma(W_x x_i(k)) \in \mathbb{R}^H. \quad (8)$$

$$\mathbf{g}_{ij}(k) = \sigma(W_{e2} \sigma(W_{e1} z_{ij}(k))) \in \mathbb{R}^H, \quad z_{ij}(k) = [M_{ij}(k), \tau_{ij}(k)]^\top. \quad (9)$$

$$\mathbf{z}_{ij}^{(0)}(k) = \mathbf{g}_{ij}(k) \odot \mathbf{h}_i(k) \in \mathbb{R}^H. \quad (10)$$

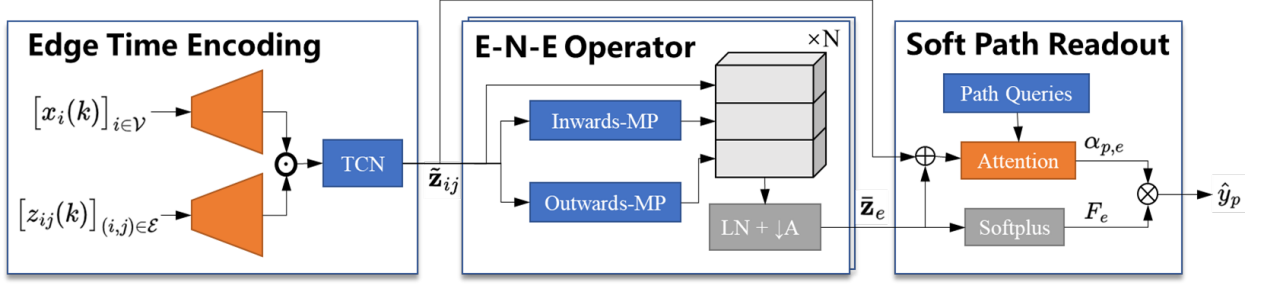


Fig. 3. Model Architecture. TCN: Time Convolution; MP: Message Passing; LN+↓A: Layer Normalize + Average Down-sample.

To model multi-step dynamics without recurrence, we apply a dilated causal 1-D convolution to the sequence $\{\mathbf{z}_{ij}^{(0)}(k)\}_{k \in \mathcal{W}_i}$:

$$\tilde{\mathbf{z}}_{ij} = \sum_{\tau=0}^{K_i-1} W_{\tau} \mathbf{z}_{ij}^{(0)}(k - d\tau), \quad k \in \mathcal{W}_i, \quad (11)$$

optionally stacked with residual connections and increasing dilation d . The output $\tilde{\mathbf{z}}_{ij} \in \mathbb{R}^H$ serves as the input to subsequent E→N→E aggregation and soft-path readout.

Edge-to-Edge Interaction on the Original Graph

To enable information exchange between edges that share an endpoint, we apply an E→N→E operator at each time step, following the Graph Network (GN) update / aggregate scheme in the message-passing paradigm (Battaglia et al., 2018; Gilmer et al., 2017). Let $e = (i \rightarrow j)$ and let $\tilde{\mathbf{z}}_e \in \mathbb{R}^H$ be the edge representation from the edge–time encoder.

We first form linear messages on edges and aggregate them at destination (in) and source (out) nodes with degree normalization:

$$\mathbf{m}_e^{in} = W_{in} \tilde{\mathbf{z}}_e, \quad \mathbf{m}_e^{out} = W_{out} \tilde{\mathbf{z}}_e. \quad (12)$$

$$\mathbf{h}_j^{in} = \frac{1}{d_j^-} \sum_{e: dst(e)=j} \mathbf{m}_e^{in}, \quad \mathbf{h}_i^{out} = \frac{1}{d_i^+} \sum_{e: src(e)=i} \mathbf{m}_e^{out}. \quad (13)$$

Each edge is updated by concatenating its own state with the source–node outgoing and destination–node incoming contexts, followed by residual and layer normalization:

$$\Delta \mathbf{z}_e = \text{MLP}([\tilde{\mathbf{z}}_e; \mathbf{h}_i^{out}; \mathbf{h}_j^{in}]), \quad \hat{\mathbf{z}}_e = \text{LN}(\tilde{\mathbf{z}}_e + \Delta \mathbf{z}_e). \quad (14)$$

After stacking L blocks across time steps, we summarize the window by simple averaging (or another permutation-invariant reducer):

$$\bar{\mathbf{z}}_e = \frac{1}{|\mathcal{W}_l|} \sum_{k \in \mathcal{W}_l} \hat{\mathbf{z}}_e(k) \in \mathbb{R}^H. \quad (15)$$

Each $E \rightarrow N \rightarrow E$ block costs $O(EH)$ per time step; stacking L layers scales linearly in L . The node reductions in (13) are implemented by segmented reductions (e.g., `scatter_add` / `scatter_mean`) for efficiency on sparse graphs.

Soft Path Readout

We obtain a fixed-size path-flow vector by attending from P learnable queries $\{\mathbf{q}_p\}_{p=1}^P$ to the *edge set*. Let $\tilde{\mathbf{z}}_e \in \mathbb{R}^H$ be the window-level edge embedding and let $\tilde{\mathbf{z}}_e = \text{Smooth}(\tilde{\mathbf{z}}_e)$ be an optional topology-aware smoothing.

Each query produces scores over edges and a normalized attention:

$$S_{p,e} = \mathbf{q}_p^\top (\tilde{\mathbf{z}}_e + \tilde{\mathbf{z}}_e), \quad \alpha_{p,e} = \frac{\exp(S_{p,e})}{\sum_{e'} \exp(S_{p,e'})}. \quad (16)$$

A nonnegative edge-level head maps each edge embedding to a scalar and aggregates by attention:

$$F_e = \text{softplus}(\mathbf{w}^\top \tilde{\mathbf{z}}_e) \geq 0, \quad \hat{y}_p = \sum_{e \in \mathcal{E}} \alpha_{p,e} F_e, \quad p = 1, \dots, P. \quad (17)$$

The attention vector $\alpha_p = (\alpha_{p,e})_{e \in \mathcal{E}}$ constitutes a *soft path* over edges, yielding an interpretable heatmap for each output dimension while providing a permutation-invariant, fixed-length readout of the edge set.

3.4. Regularization

We minimize a physics-aware objective tailored to AVI→path-flow regression. The data-fit term uses mean squared error, the maximum-likelihood estimator under homoscedastic Gaussian noise, while two complementary regularizers address identifiability and spatial coherence of “soft paths” on the detector graph.

$$\mathcal{L}_{\text{mse}} = \frac{1}{P} \sum_{p=1}^P (\hat{y}_p(t) - y_p(t))^2. \quad (20)$$

The mapping from AVI observations to path flows is underdetermined: many edge combinations can explain the same windowed signal. Without a prior, the attention over edges for path p may collapse onto a few edges, producing brittle and poorly calibrated solutions. We therefore penalize low-entropy attentions by minimizing the KL divergence to the uniform distribution, which is equivalent (up to a constant) to *maximizing* Shannon entropy and has been shown to act as a strong confidence penalty in deep networks:

$$\mathcal{L}_{\text{ent}} = \frac{1}{P} \sum_{p=1}^P \text{KL}(\alpha_p \| \mathbf{u}) = \frac{1}{P} \sum_{p=1}^P \sum_{e \in \mathcal{E}} \alpha_{p,e} \log(\alpha_{p,e} |\mathcal{E}|). \quad (21)$$

This term promotes non-brittle, multi-edge soft paths, improves identifiability, and mitigates the empirically observed entropy collapse in attention layers.

To encode corridor continuity on the detector graph, we impose a Laplacian quadratic penalty on each attention vector. Let A_e be the (symmetrized) edge-adjacency matrix (edges are neighbors if they share a node), $L_e = D_e - A_e$

its Laplacian, and $\alpha_p = (\alpha_{p,e})_{e \in \mathcal{E}}$ the attention for path p . The Dirichlet energy

$$\mathcal{L}_{\text{smooth}} = \frac{1}{P} \sum_{p=1}^P \alpha_p^\top L_e \alpha_p. \quad (22)$$

penalizes rapid variation across adjacent edges (i.e., $\sum_{(e,e') \in \mathcal{N}} A_{e,e'} (\alpha_{p,e} - \alpha_{p,e'})^2$), encouraging spatially coherent soft paths aligned with feasible travel corridors and reducing spurious oscillations.

The final objective is a weighted sum

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}} + \lambda_{\text{sm}} \mathcal{L}_{\text{smooth}}, \quad (23)$$

with $\lambda_{\text{ent}}, \lambda_{\text{sm}} > 0$ tuned on a chronological validation split. Both regularizers are scale-free, differentiable, and complementary: $\mathcal{L}_{\text{ent}}$ combats ill-posedness by dispersing probability mass over plausible edges, while $\mathcal{L}_{\text{smooth}}$ injects a graph-physical prior that favors contiguous, corridor-like attention patterns.

4. Experiments and Validation

4.1. Experimental Setup

We evaluate **PFQ-GNN** on the Taicang urban network (Jiangsu, China). A faithful replica is built in SUMO (Fig. 4), comprising 4,437 nodes and 10,376 directed links spanning expressways, arterials, and locals. AVI detectors are installed at intersection approaches and key mid-links; each detector logs vehicle ID, type, and timestamp for all passing vehicles, covering 1,735 links. One week of heavy-duty truck GPS trajectories (real-time positioning is mandated for trucks >12 t) and citywide AVI records are used to construct graph inputs and supervision signals.

To form OD–path candidates, GPS-derived OD points are embedded in $\mathbb{R}^4$ via multidimensional scaling and clustered by k -means (512 heavy-duty OD pairs). Freight-related POIs (factories, ports, logistics parks) are retrieved from Amap to construct additional OD candidates for light-duty trucks. For both classes, a link-penalization search yields candidate paths; prior path flows are aggregated using a path-overlap similarity.

Because multi-day samples are necessary for spatio-temporal learning, we synthesize daily demand patterns by injecting zero-mean Gaussian noise (10%–30%) into the *hourly* prior path flows, and then load them into SUMO. the simulator returns synthetic AVI observations under the same detector layout.

Each training sample covers a *4-hour* window, discretized into $T=4$ one-hour steps (consistent with $\Delta t \gg \tau_e$). For each step we record (i) detector counts, (ii) matched flows between detector pairs, and (iii) matched travel times. To increase information density while controlling graph size, the 256 most active detectors are retained to form the AVI graph; the induced subgraph is connected (edge count E and mean degree are reported in the appendix). After preprocessing, we obtain 4,200 spatio-temporal graphs (256 nodes, $T=4$) paired with a path-flow label $\mathbf{y} \in \mathbb{R}^{2 \times 100}$ (100 heavy-duty and 100 light-duty paths).

We train PFQ-GNN with a 2-layer temporal convolutional module (edge-wise TCN), followed by three stacked $E \rightarrow N \rightarrow E$ blocks for on-graph edge–edge interaction; residual connections stabilize training and enlarge the receptive field. The hidden size is 64 for both node and edge embeddings (129,025 trainable parameters). Optimization uses **Adam** (learning rate 3×10^{-4} , weight decay 10^{-3} , $\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$), with batch size 32, 500 epochs, mixed precision (PyTorch AMP), and global-norm gradient clipping at 10. Training adopts $K=5$ -fold cross-validation. The supervised objective is $\mathcal{L} = \mathcal{L}_{\text{mse}} + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}} + \lambda_{\text{sm}} \mathcal{L}_{\text{smooth}}$ with fixed weights $\lambda_{\text{ent}} = 10^{-4}$ and $\lambda_{\text{sm}} = 5 \times 10^{-4}$. All experiments run on an Intel i9-13900K CPU, an NVIDIA RTX 4090 (24 GB), and 128 GB RAM.



Fig. 4. (a) Road Network in SUMO; (b) Truck Flow Distribution

4.2. Metrics

To respect temporal dependence, we adopt *rolling-origin* time-series cross-validation with five forecast origins; training windows strictly precede validation/test windows, and scores are averaged across origins. We additionally conduct a *spatial block* CV sensitivity by holding out contiguous detector subgraphs to probe robustness under spatial dependence. Both designs mitigate temporal/spatial leakage that can bias random K -fold splits.

Given count-like targets with strong class imbalance across paths, we report *weighted* errors so that large-flow paths contribute proportionally more. Let $y_{k,r}$ and $\hat{y}_{k,r}$ denote the actual and predicted flows for path r at step k , and define weights $w_{k,r} = y_{k,r} \geq 0$. Units for WMAE/WRMSE are *vehicles per hour*. We use

$$\text{WMAE} = \frac{\sum_{k=1}^K \sum_{r=1}^R w_{k,r} |y_{k,r} - \hat{y}_{k,r}|}{\sum_{k=1}^K \sum_{r=1}^R w_{k,r}}, \quad \text{WRMSE} = \sqrt{\frac{\sum_{k=1}^K \sum_{r=1}^R w_{k,r} (y_{k,r} - \hat{y}_{k,r})^2}{\sum_{k=1}^K \sum_{r=1}^R w_{k,r}}}, \quad (10)$$

and their percentage analogues

$$\text{WMAPE} = \frac{\sum_{k=1}^K \sum_{r=1}^R |y_{k,r} - \hat{y}_{k,r}|}{\sum_{k=1}^K \sum_{r=1}^R y_{k,r}}, \quad \text{WMSPE} = \frac{\sum_{k=1}^K \sum_{r=1}^R (y_{k,r} - \hat{y}_{k,r})^2}{\sum_{k=1}^K \sum_{r=1}^R y_{k,r}^2}. \quad (11)$$

4.3. Main Results

We compare this model with the Res3D model [Tang et al. \(2021\)](#). To investigate both predictive accuracy and training stability, we analyze the regression plots of predicted vs. observed path flows (Fig. 5) and the training–validation loss curves (Fig. 6). Several observations can be made:

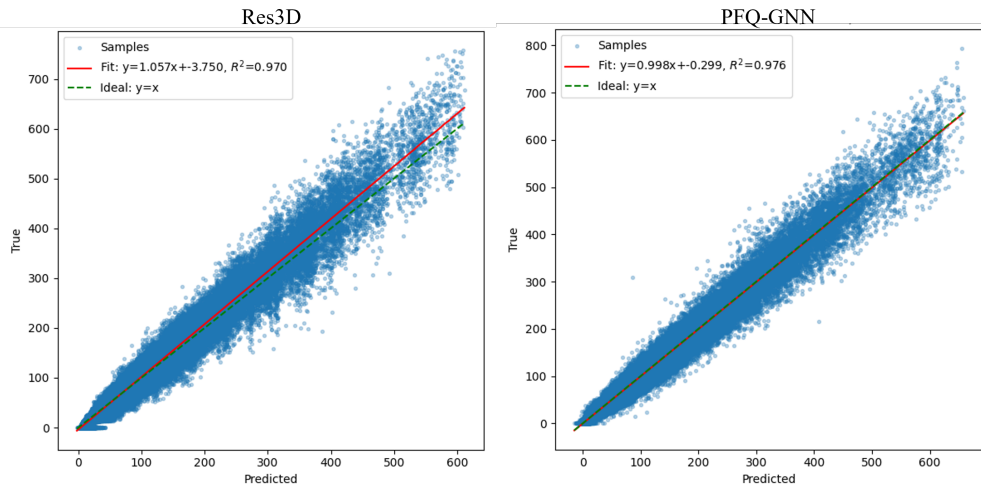


Fig. 5. True vs predicted path flows: Res3D vs PFQ-GNN.

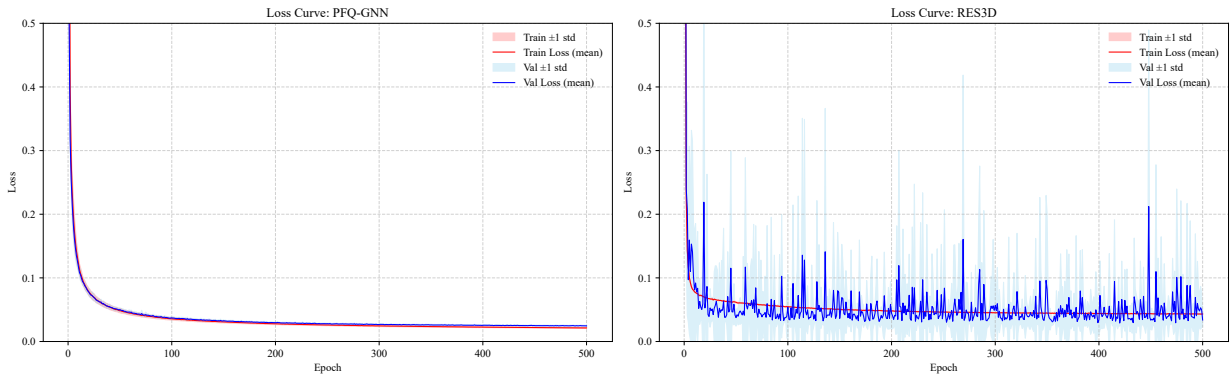


Fig. 6. Training–validation loss curves: PFQ-GNN (left) vs Res3D (right). Shaded areas denote ± 1 standard deviation across folds.

- **Fast and smooth convergence.** PFQ-GNN demonstrates rapid convergence within the first 50 epochs, with both training and validation losses decreasing steadily. Around epoch 100, the curves plateau at a low level and remain stable throughout the remaining epochs. The variance band across folds is narrow, indicating low sensitivity to initialization and data partition.
- **Instability in Res3D.** By contrast, Res3D exhibits strong oscillations in validation loss across the entire training horizon, particularly between 50 and 300 epochs. Even when training loss decreases, the validation loss fluctuates significantly, suggesting poor generalization and susceptibility to overfitting.
- **Predictive accuracy and bias.** As shown in Fig. 5, PFQ-GNN achieves a regression line closely aligned with the ideal $y = x$ reference (slope ≈ 0.998 , intercept ≈ 0.299 , $R^2 = 0.976$). In contrast, Res3D displays a steeper slope (≈ 1.057) and higher intercept (≈ 3.750 , $R^2 = 0.970$), indicating systematic bias with overprediction at high flows and underprediction at low flows. PFQ-GNN thus provides more balanced and unbiased estimates across the entire flow range.
- **Overall superiority.** Taken together, Figs. 5 and 6 highlight PFQ-GNN’s advantages over Res3D. It not only converges faster and more smoothly but also yields higher predictive accuracy with reduced variance and lower systematic bias. Such robustness reduces the need for extensive hyperparameter tuning and improves scalability to larger or more complex traffic networks.

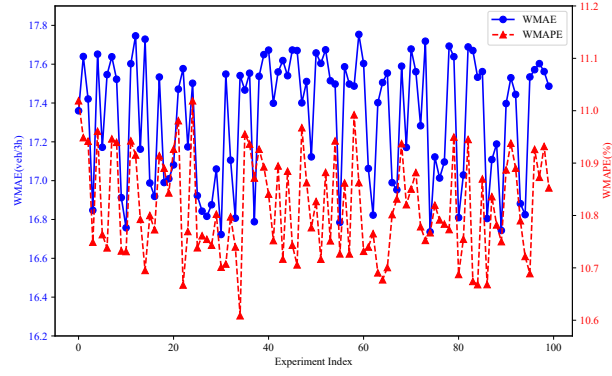


Fig. 7. PFQ-GNN metric distributions across repeats

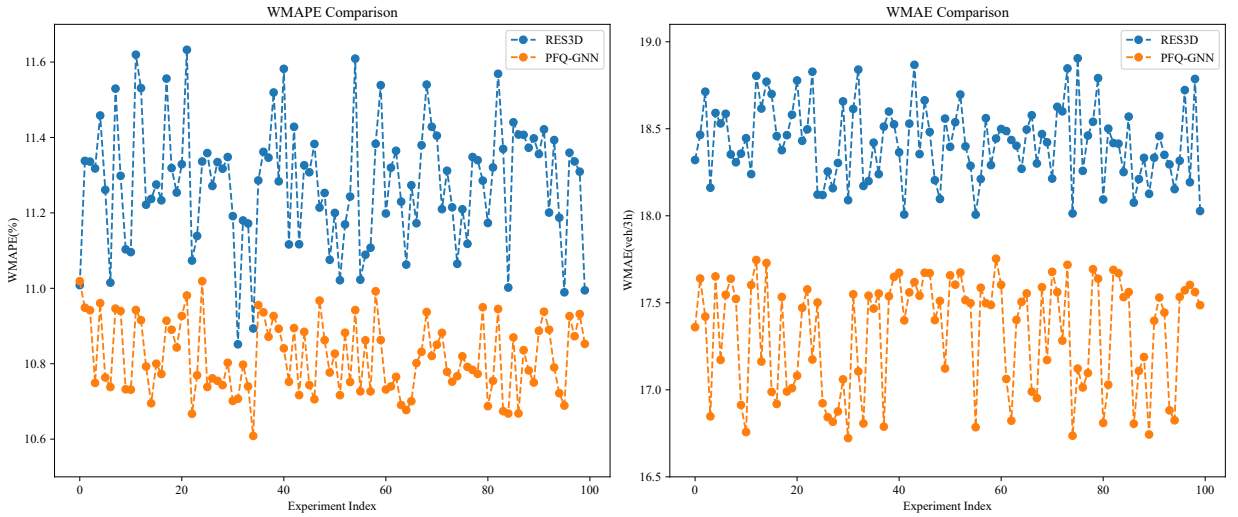


Fig. 8. PFQ-GNN vs Res3D contrast plot

Under identical splits, PFQ-GNN achieves lower errors with smaller variance than Res3D. WMAPE (mean $\pm$ sd) decreases from 11.2791 ± 0.1621 (Res3D) to 10.8175 ± 0.0963 (PFQ-GNN), an absolute improvement of 0.462 pp (4.09% relative) with a 41 reduction in variance. Likewise, WMAE drops from 18.424 to 17.326 (veh/h, 6.0%); WRMSE from 27.129 to 25.479 (6.1%); and WMSPE from 1.493 to 1.356 (9.2%). Across the 100 repeated runs (5-fold $\times$ 20 repeats; Fig. 7), PFQ-GNN's curves lie consistently below those of Res3D without drift (Fig. 8), indicating stability with respect to resampling and initialization.

For statistical confirmation, we compare paired WMAPE per split: the Wilcoxon signed-rank test yields statistic 1.0 with $p = 4.02 \times 10^{-18}$, rejecting the null of no difference. The Nadeau–Bengio corrected t -test (accounting for dependence under repeated CV) gives $t = 6.528$, $p = 2.88 \times 10^{-9}$ with a 95% CI of (0.321, 0.602) pp. The Cliff's $\delta = 0.985$ indicates a large effect, i.e., PFQ-GNN outperforms Res3D in $\sim 99.25\%$ of paired runs.

4.4. Ablation Study

Table 1 summarizes the ablation study across different architectural variants. We refine the model names for clarity: the *3D Residual CNN Baseline* (Res3D) is adopted from prior work; *Edge-Only GNN (No Temporal Modeling)* removes the temporal convolutional module entirely; *Shallow Edge-Only GNN (1-Layer)* retains only a single E $\rightarrow$ N $\rightarrow$ E

block without temporal encoding; *Shallow PFQ-GNN + Regularization* includes a minimal temporal convolution with additional entropy and Laplacian regularization; and finally, the full *PFQ-GNN (Ours)* integrates all components.

The results reveal several key findings. First, removing temporal modeling (Edge-Only GNN) leads to a sharp performance drop (WMAPE 12.62%), confirming the critical role of temporal dynamics in OD–path flow prediction. Simply deepening the E→N→E stack without temporal encoding further increases error and variance, suggesting that spatial propagation alone cannot substitute for temporal context. Second, adding regularization improves robustness but still falls short of Res3D. Third, the complete PFQ-GNN consistently outperforms all baselines: it reduces WMAPE to 10.82% while maintaining a lightweight parameter size (129K), compared to the much larger Res3D baseline (864K). Finally, PFQ-GNN exhibits the smallest variance across runs, indicating stable convergence and strong generalization. These results highlight that carefully designed edge-temporal interaction and physics-aware regularization contribute more to performance and efficiency than brute-force parameter scaling.

Table 1. Ablation on temporal modules, edge–node–edge depth, and regularization (mean ± std over runs).

Model	WMAE (veh/3h)	WRMSE (veh/3h)	WMAPE (%)	WMSPE	#Params
3D Residual CNN Baseline (Res3D)	18.42 (± 0.22)	27.13 (± 0.37)	11.28 (± 0.16)	1.49 (± 0.02)	864K
Edge-Only GNN (No Temporal Modeling)	19.39 (± 0.24)	28.09 (± 0.33)	12.62 (± 0.21)	1.74 (± 0.04)	117K
Shallow Edge-Only GNN (1-Layer)	20.61 (± 0.53)	29.90 (± 1.04)	13.45 (± 0.18)	1.97 (± 0.08)	67K
PFQ-GNN (MSE Loss without Regularization)	18.98 (± 0.39)	27.63 (± 0.58)	12.23 (± 0.23)	1.66 (± 0.06)	129K
Full PFQ-GNN	17.33 (± 0.32)	25.48 (± 0.61)	10.82 (± 0.10)	1.36 (± 0.04)	129K

5. Conclusion and future work

This paper develops a new spatio–temporal graph neural network, PFQ-GNN, that maps AVI detector streams on a directed detector graph to path-level freight flows. The model integrates an edge-wise temporal encoder (dilated TCN), an on-graph E→N→E interaction for edge-to-edge communication without line-graph expansion, and a query-based soft-path readout regularized by attention entropy and edge-adjacency Laplacian priors. A calibrated data pipeline fusing GPS/AVI/POI information and a rolling-origin evaluation protocol ensure operational relevance and statistical rigor.

In conclusion, the main contributions of the presented work are as follows:

- **Framework innovation.** PFQ-GNN provides a compact and interpretable architecture that directly maps detector data to path-level flows, combining edge-wise temporal encoding with efficient on-graph edge-to-edge propagation. This design overcomes the limitations of conventional link-based or enumerated-path methods and demonstrates the feasibility of scalable path flow estimation in large networks.
- **Empirical performance.** On a real urban freight network with low AVI penetration and biased, low-sampling GPS data, PFQ-GNN consistently outperforms a strong 3D-CNN baseline in both weighted absolute and percentage errors, while exhibiting markedly lower variance across 5-fold×20 repeated splits. These improvements highlight the robustness and generalizability of the proposed framework under realistic sensing constraints.
- **Freight travel insights and management implications.** The entropy- and Laplacian-regularized soft-path attention produces corridor-level patterns that align with observed truck flows, revealing dominant freight corridors, temporal variability, and heterogeneous route choices. These interpretable outputs provide an analytical basis for corridor management, demand regulation, and infrastructure planning, bridging advanced graph learning with evidence-based freight policy.

Despite promising results, several drawbacks of this work could be improved: the reliance on simulation-augmented supervision may introduce bias; detector selection (top-256 by activity) can limit coverage; and uncertainty estimates are not yet attached to path-flow outputs.

For future research, we will continue to (i) couple learning with differentiable assignment and systematic back-validation to independent AVI aggregates, (ii) develop uncertainty-aware prediction (e.g., distribution-free intervals) for risk-sensitive operations, (iii) extend to dynamic topologies and multi-resolution pooling for scalability and cross-city transfer, and (iv) fuse additional freight data sources (ETC/OBU, e-waybills, IoT) to strengthen multimodal representations and deployment readiness.

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